In-House Practical Training
on
Clustering different types of words in silent speech patterns in Brain-Computer Interfaces (BCIs) using ML

Submitted to
Amity University Uttar Pradesh



in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology
in
Computer Science & Engineering

Submitted by

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I, **Arpita** , student of B. Tech(CS&E) 5th Semester hereby declare that the In-House Practical Training titled **"Clustering different types of words in silent speech patterns in Brain-Computer Interfaces (BCIs) using ML"** which is submitted by me to Department of Computer Science and Engineering, Amity School of Engineering and Technology, Lucknow, Amity University Uttar Pradesh, Lucknow Campus, in partial fulfillment of requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering has not been previously formed the basis for the award of any degree, diploma or other similar title or recognition.

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CERTIFICATE

On the basis of declaration submitted by **Arpita** student of B.Tech (Computer Science and Engineering) 5th semester , I hereby certify that the In-House Practical Training titled **"Clustering different types of words in silent speech patterns in Brain-Computer Interfaces (BCIs) using ML"** which is submitted to Amity School of Engineering and Technology, Amity University Uttar Pradesh, Lucknow, in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering , is an original contribution with existing knowledge and faithful record of work carried out by her under my guidance and supervision.

To the best of my knowledge this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere.

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ABSTRACT

Especially for those who have misplaced their capacity to talk, Brain-Computer Interfacing (BCIs) for noiseless discourse are getting to be increasingly critical as substitute shapes of communication. This consider analyzes whether machine learning can be utilized to genuinely gather words, counting things, verbs, and work words, and how these word sorts are spoken to within the brain amid quiet (envisioned) discourse. To explore basic designs in neural signals, I utilize clustering methods on non-invasive EEG information gotten amid envisioned discourse assignments. My objective is to decide whether, indeed within the nonattendance of capable of being heard discourse, words from different etymological categories create perceivable brain action.

I finish this by utilizing both self-supervised and unsupervised learning procedures, which permit the demonstrate to memorize reliable highlights over subjects and sessions without intensely depending on labeled information. In arrange to upgrade generalizability, I too handle the issue of EEG flag inconstancy by executing cross-session investigation. The coming about clusters are surveyed agreeing to their consistency among members and their arrangement with etymological categories.

The comes about demonstrate empowering designs that infer particular word sorts might, in reality, evoke unmistakable neural responses. This might have a huge affect on the creation of more common and compelling BCIs for quiet discourse, empowering clients to precise whole words or indeed sentences fair with their considerations. The extreme objective of this work is to development the advancement of valuable, brain-based communication frameworks for the underprivileged.

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1. INTRODUCTION

The inner handle of envisioning or getting prepared to talk without really talking is known as quiet discourse. The potential employments of quiet discourse inquire about in assistive communication advances, especially for individuals who are incapable to talk since of physical or neurological impedances, have drawn consideration in later a long time. Noiseless discourse is troublesome to interpret utilizing standard strategies since it needs unmistakable or capable of being heard prompts, in differentiate to conventional discourse. Brain-Computer Interfacing (BCIs) are valuable in this circumstance. In arrange to encourage communication between the brain and outside gadgets, brain-computer interfacing (BCIs) translate neural movement. When it comes to noiseless discourse, brain-computer interfacing (BCIs) utilize electroencephalography (EEG) to translate inside produced discourse signals, such as envisioned words or expressions, specifically from brain action.

Exact word-level acknowledgment in noiseless discourse is still exceptionally troublesome, in spite of later advancements in machine learning making strides flag interpreting at the phoneme or character level. The changeability of EEG signals makes it troublesome for current BCI strategies to generalize over sessions and subjects. Moreover, they treat all inputs as partitioned substances and barely ever take into consideration the semantic or phonetic structure of words. In arrange to fill these crevices, this consider explores how translating models can be improved by gathering related word types—such as things, verbs, and work words—based on their neural representations. Clustering can offer assistance machine learning models way better capture word-level designs and reveal inactive structure in brain signals.

Exploring whether words from different etymological categories cause diverse brain action amid noiseless discourse and whether unsupervised or self-supervised clustering methods can effectively gather these words is the most objective of this ponder. The comes about of this think about may offer assistance make BCI frameworks for quiet discourse communication that are more exact, dependable, and simple to utilize.

2. LITERATURE REVIEW

2.1. Overview of Silent Speech and Brain-Computer Interfaces (BCIs)

Silent speech, also called covert or imagined speech, is when someone thinks about speaking words or sentences without actually making any sounds or moving their mouth. It's a way people mentally practice or imagine talking. This idea has become a big focus in neuroscience and biomedical engineering because it could help people who can't speak due to conditions like ALS, brainstem strokes, or having their voice box removed. Creating tools that can understand silent speech could help these people communicate better and improve their daily lives.

Brain-Computer Interfaces, or BCIs, are one way to help with silent speech. These systems let the brain talk directly to devices without using the usual body signals. They work by reading brain activity, often through EEG, which measures electrical signals on the scalp. For silent speech, BCIs pick up on the brain's signals when someone is imagining words and then translate those into speech sounds or words.

EEG-based brain-computer interfaces are popular because they are easy to carry, not too expensive, and can record brain activity very quickly over time. But it's difficult to understand silent speech using EEG signals. These signals are weak compared to the background noise and can be messed up by things like blinking eyes, muscle movements, and outside distractions. Also, EEG doesn't show very detailed brain activity maps, making it hard to exactly find where in the brain certain speech-related actions happen.

Even though there are some challenges, new studies have shown hopeful results in understanding speech that is imagined using machine learning and advanced ways to process signals. Researchers have looked at different parts of language like vowels, individual sounds, words, and short sentences, and they're getting better at recognizing them accurately and applying the findings to different situations. The use of deep learning and self-teaching methods in AI has sped up progress, helping create stronger models that can work well even with limited or unclear data.

Combining silent speech recognition with brain-computer interfaces is helping create communication tools that are easy to use, don't require surgery, and can read a person's thoughts without them speaking or making facial movements.

These developments could greatly help in rehabilitating people with neurological issues, improve communication for those who can't speak, and even make it easier to control devices without using hands in everyday tech interactions.

- Brigham & Kumar (2010) introduced imagined speech decoding via EEG, setting early precedent.

- Bocquelet et al. (2016) and Herff & Schultz (2016) reviewed covert speech BCI designs, highlighting the challenge of translating neural activity into speech-specific control signals

2.2. EEG-Based Speech Recognition Studies

Electroencephalography (EEG)-based speech recognition has become an important area of study within Brain-Computer Interfaces (BCIs). The goal is to understand and translate brain signals that are related to speech. Unlike regular speech recognition systems that use sounds from the environment, EEG-based methods read speech-related brain activity directly. This makes them useful for silent speech interfaces (SSIs), especially for people who can't speak because of injury or illness.

Early research in this area focused on identifying simple speech sounds or telling the difference between two words in the mind.

For example, Suppes and his team in 1997 showed that EEG could pick up different brain patterns for different English words, which was a big step forward. Later studies improved this by using techniques like linear discriminant analysis (LDA), support vector machines (SVM), and eventually deep learning models to make speech recognition more accurate for many people.

A lot of research has looked at how to classify vowels and phonemes using EEG data. Studies like those by Martin et al. (2014) and Dash et al. (2020) found good results in identifying imagined vowels like /a/, /i/, and /u/, as well as other speech sounds. However, these studies used small sets of words and had some differences in how people responded. These results show that it's possible to understand speech from brain signals without surgery, and they encouraged more research into bigger speech units.

Recent studies are focusing more on understanding whole words and short phrases, expanding what can be achieved with EEG-based decoding. Nguyen and their team in 2017 used deep belief networks to classify imagined words, and they had some success. Cooney and others in 2020 used convolutional neural networks and transfer learning to decode EEG signals for different words and speech commands. These studies show how important it is to learn both the spatial and time-based features of EEG signals, and how using advanced neural network structures can help deal with the changes and noise in EEG data.

At the same time, the ability of systems to work across different people and recording sessions has become a major area of focus. Even if models do well with the data they are trained on, EEG signals can change a lot from one person to another or between different recording times, which makes it hard to use them in real life. To deal with this, researchers have suggested methods like adapting to different data sources, extracting features that stay the same across sessions, and training models that work for anyone (Lawhern et al. , 2018; Tian & Liu, 2022).

Another important trend is the use of mixed BCI systems that combine EEG with other types of data, such as muscle signals (EMG) or blood flow measurements (fNIRS).
Even though these systems are more complicated, they have been found to improve the accuracy and reliability of results, especially when used for recognizing speech through multiple types of signals.
Even though there has been a lot of progress, EEG-based speech recognition is still not as accurate as older methods that use sound. However, the ability to understand speech that is imagined or said silently by reading brain signals is a big change in how we communicate. This is especially important for people who can't move or speak, like those with locked-in syndrome or serious muscle and nerve disorders. The area is growing quickly, and new methods such as self-supervised learning, attention mechanisms, and transformers are being tested to make the system work better and more reliably with EEG data.

- Suppes et al. (1997) demonstrated phoneme and word classification using EEG with traditional classifiers.
- Porbadnigk et al. (2009) examined temporal dynamics in EEG-based speech recognition.
- Herff et al. (2016–2020) and Martin et al. (2018) explored ECoG and EEG for imagined speech, highlighting frequency and spatial correlations.
- Nguyen et al. (2017) used deep belief nets for word classification.
- Cooney et al. (2020) introduced CNNs and transfer learning to improve recognition.
- Kim et al. (2023) applied diffusion-based autoencoders for EEG decoding at Interspeech 2023

2.3. Clustering and Unsupervised Learning in BCI

Clustering and other unsupervised learning methods are becoming more important in Brain-Computer Interface (BCI) research, especially when there isn't enough labeled data. Traditional BCI approaches, like those that use imagined or silent speech, often face problems because it's hard to collect enough data, people get tired, and it's unclear what the labels should be. In such cases, unsupervised learning helps by finding patterns in raw or lightly processed EEG signals without needing labels.

Algorithms like K-Means, DBSCAN, and hierarchical clustering are commonly used with EEG data in BCI projects.

These include tasks like classifying motor imagery, detecting mental states, recognizing emotions, and even decoding silent speech. The main goal of clustering in BCI is to group similar brain activity patterns together, even when we don't know what those patterns mean beforehand. This helps uncover hidden structures in the data that might represent different mental tasks, imagined words, or categories of meaning.

In the area of silent speech, grouping similar data points helps find clear patterns that represent different words in brain signals. Unlike methods that use already labeled words to train models, grouping lets researchers discover how the brain represents words without needing predefined labels. For example, some studies have looked into whether the brain creates separate brain signal patterns for different types of words, like nouns, verbs, and function words. These studies usually start by using techniques like power spectral density, band power, wavelet changes, or hidden patterns from autoencoders to extract useful features before applying grouping methods.

Some studies have used K-Means clustering on EEG data collected when people imagine speaking to see if different groups of brain activity patterns match up with different spoken or imagined words (Nguyen et al. , 2019). The results showed there were some groupings, but they also pointed out that usual methods for measuring distance between data points don't work well in high-dimensional EEG data, especially when the data is messy or the features are similar. To fix these problems, researchers have tried more advanced clustering techniques, like DBSCAN, which can find clusters of any shape and deals with noisy data, and spectral clustering, which uses graph-based methods to find similar patterns.

Recent advances in using deep learning along with clustering methods, like deep embedded clustering (DEC) and self-supervised learning, have greatly improved the field. These models learn both the features of data and how to group them at the same time, which works well for EEG signals that have detailed and hard-to-catch patterns. Research using auto encoders to create a hidden space for clustering has shown good results in grouping brain responses based on the meaning or sound of words. This opens up new possibilities for creating silent speech brain-computer interfaces (BCIs) that don't need lots of manual labels.

Clustering has also been used in subject adaptation and transfer learning. In this approach, the brain activity from one person helps group or cluster data from another person. The idea is to create models that work better across different people. For example, when models are trained using data that has been grouped together from multiple people, they perform better when trying to classify brain signals from new people. This helps deal with the challenge of differences between people, which is a big problem in EEG-based brain-computer interfaces.

Another important part is checking how good the clusters are and how well they match the actual language labels. Tools like the silhouette score, Davies–Bouldin index, and cluster purity help measure how clear and separate the groups are. When we know the correct answers, like the categories of words,

we can see how well the clusters match those labels. This helps us understand how well the brain separates different imagined words.

In short, clustering and unsupervised learning have created new ways to understand silent speech from brain signals.

They help find hidden patterns in the brain, make it easier to use data from one person on another, and offer a way to label data with less information. These improvements are key for the future of brain-computer interfaces, which want to enable fast, real-time communication using only brain activity.

- Early methods applied K-Means, DBSCAN, spectral clustering to EEG for motor imagery, and extended to imagined speech (Nguyen et al., 2019).

- Improved deep clustering models like DEC and DEC-style autoencoders emerged to jointly optimize representations and clusters.

- Zhang et al. (2020) introduced clustering-based multi-task feature learning for EEG, leveraging subclass discovery to improve decoding accuracy

2.4. Feature Extraction and Preprocessing Techniques

Good feature extraction and data preparation are key to making any machine learning system work well when dealing with EEG signals, especially for understanding silent speech. EEG signals are tricky because they change over time, have a lot of data points, and are often mixed with noise from the body and the environment. Because of this, it's important to have a clear and thoughtful plan for cleaning the data and creating useful features. This helps in focusing on the important brain activity related to the task and makes the models used later more reliable and accurate.

Preprocessing EEG Data

The preprocessing process usually starts by removing unwanted noise caused by things like eye movements, muscle activity, heartbeats, and electrical interference. One common method used for this is Independent Component Analysis (ICA), which helps separate mixed signals into parts that are statistically independent, making it easier to remove non-brain-related noise.

Another important step is band-pass filtering, which helps focus on specific frequency ranges of the EEG signal that are most useful for speech-related tasks.

For imagined or silent speech, researchers often look at frequencies like alpha (8–13 Hz), beta (13–30 Hz), and gamma (>30 Hz), which are linked to language, movement imagery, and thinking. The filters used usually cover a range from 0. 5 to 40 Hz, depending on the study and the quality of the signal.

Baseline correction and channel interpolation might also be used, especially when working with multiple sessions or subjects, to keep the data sets consistent. Also, down-sampling is usually done to make the processing faster without losing too much of the important neural information.

Feature Extraction Techniques

After preparing the data, the next step is to extract features from the raw EEG signals. This process turns the signals into numbers that can be used by machine learning models. These features help in understanding the time-based, frequency-based, location-based, or complex patterns of brain activity linked to silent speech.

- Time-domain features are basic stats like average, spread, skew, kurtosis, and RMS of the EEG signal in short time periods. These are simple but often helpful for telling different types of signals apart.

- Frequency-domain features are made using FFT or Welsh's method to find power spectral density.This shows how much power is in different frequency ranges, which can tell us which brain wave patterns are active during imagined speech.

- Time-frequency features come from STFT or Wavelet Transform.They show how the signal changes over time and what frequencies are present at each moment. Wavelet methods are especially good for EEG because they can look at different time scales, making them better for signals that change over time.

- Spatial features like Common Spatial Patterns (CSP) are commonly used in motor imagery brain-computer interfaces and are now being used more for speech decoding. CSP finds spatial filters that increase the difference in variance between different conditions, which helps separate classes better.
- Connectivity-based features look at how different brain channels interact, using measures like phase locking value (PLV) or coherence. These metrics show how brain regions work together in sync during internal speech.

- Nonlinear features help understand complex patterns and changes in signals, including entropy measures like Sample Entropy and Spectral Entropy, fractal dimension, and Higuchi’s dimension.These features

  have been helpful in telling apart different mental states during imagined speech tasks.

In recent years, there has been a move towards using representation learning in deep learning, which helps models like auto-encoders, convolutional neural networks (CNNs), and recurrent neural networks (RNNs) automatically find important features from raw or slightly processed EEG data. These features, which are learned by the models, usually work better than features that are created manually, especially when used with unsupervised learning methods for grouping data.

Challenges and Considerations

Even though there are many methods available, extracting features from EEG data is still difficult. Differences between people, the fact that results can't always be repeated, and problems with over-fitting because of large data sets are big issues. To deal with this, people often use techniques like mutual information, recursive feature elimination, or principal component analysis to cut down on the amount of data and make the results better for different people.

Another thing to think about is how well these methods work in real time, especially when using brain-computer interfaces. It's important to find a good balance between how well the method can tell things apart and how much computer power it uses, so it can work quickly during live use.

- Standard preprocessing (ICA, band-pass filters 0.5–40 Hz, notch) and feature extraction (PSD, STFT, CSP, wavelets) detailed in reviews by Alzahrani et al. (2024) and IJACSA (2022) .

- The shift toward deep learning began around 2017, as shown in Frontiers' survey of deep- learning in BCI (2017–2022)

2.5. Auto-encoders and Dimensionality Reduction in EEG

Electroencephalographic (EEG) data, especially when used for silent speech decoding, has some special challenges. This is because EEG data is very complex in terms of how much information it holds, how time-based it is, and how weak the useful signal is compared to the background noise. Because of this, it's hard and not very efficient to use traditional methods for analyzing and preparing features from this type of data. So, dimensionality reduction techniques have become very important in EEG research. These techniques help make the data simpler to work with while keeping the most important and useful parts. One popular method is auto-encoders, which are a kind of deep neural network. They are especially useful because they can automatically learn and create compact, meaningful features from the data without needing labeled examples.

Traditional Dimensionality Reduction Approaches

Before deep learning became widely used, people often used techniques like Principal Component Analysis (PCA), Independent Component Analysis (ICA), and Linear Discriminant Analysis (LDA) to simplify EEG data. PCA is popular because it's easy to use and good at keeping important information related to variance, which helps with reducing noise and making data easier to visualize. ICA has been widely used to remove unwanted signals and separate different sources in EEG data from multiple channels.

Although these methods are computationally efficient, they don't work well for capturing the complex and nonlinear patterns in EEG signals that happen during imagined speech.

Because of this, there's been a growing move towards using nonlinear, data-driven approaches like auto-encoders.

Autoencoders in EEG Feature Learning

Auto-encoders are a type of neural network that learns to recreate its input at the output layer. They do this by first compressing the input into a simpler form, called the bottleneck layer. This compressed version helps reduce the complexity of the data while keeping the most important information. In EEG studies, auto-encoders are used on raw or cleaned EEG data to create embedding that are specific to each person or task. These embedding are good at handling noise and differences between people.

Different types of auto-encoders, like Stacked Auto-encoders, Denoising Auto-encoders, and Variational Auto-encoders, have been used effectively in brain-computer interface systems based on EEG data :

- Stacked Auto-encoders use many layers to learn features in a step-by-step way. This helps when working with big sets of EEG data from different mental states or when people imagine speaking.

- Denoising Auto-encoders are trained to fix messy signals and make them clear. This helps the model focus on important patterns and avoid being distracted by random noise, which is especially important because EEG signals often have low quality.

- Variational Auto-encoders use a method based on probability to create models of brain activity. This makes them good for showing uncertainty in how the brain works and for making new EEG data to help improve training.

Applications in Clustering and Classification

Auto-encoders are being used in a promising way for silent speech brain-computer interfaces. One key use is combining them with clustering algorithms. Studies show that the hidden features from autoencoders help group similar data better than using raw data or manually created features. For example, a method called Deep Embedded Clustering uses an autoencoder along with a clustering goal. This method works together to improve both how well data is reconstructed and how well it is grouped. This helps sort brain signals into groups that match different imagined words or types of language, like nouns and verbs.

In tasks where the goal is to classify brain activity, features from auto-encoders are used as input for models like support vector machines, random forests, and deep neural networks.

These models perform better at telling apart different mental tasks or speech-related brain states. Also, these features help models work better across

different people and sessions, which is an important challenge in real-world applications using EEG data.

Challenges and Future Directions

Auto-encoder models have some great benefits, but they need a lot of data and careful setup of their structure and settings. They can also be prone to over fitting, which is a big problem when there's not much EEG data available. Plus, it's hard to understand what the features they learn actually mean, making it tough to connect them directly to thoughts or language.

New developments are helping to fix these issues. Using methods like attention mechanisms, self-supervised learning, and temporal convolutional auto-encoders is showing good results in creating better and more understandable representations. Researchers are also looking into mixing auto-encoders with other models like LSTMs to better handle the time-based patterns in imagined speech.

In short, auto-encoders are really useful tools for reducing the amount of data in EEG signals. They help find simpler, more reliable, and more focused ways to represent information that's relevant for specific tasks. When used with methods like clustering and classification, they become an important part of building better silent speech brain-computer interfaces.

- Early hierarchical deep learning (2019) showed ≈23% performance gains over baselines

- Motor- imagery BCIs adopted KNN- enhanced auto-encoders (2019)

- Lee et al. (2021) used CNN- auto-encoders to transfer overt to imagined speech features (“DAL”)

- Kim et al. (2023) combined diffusion models and conditional auto-encoders .

2.6. Identified Gaps and Research Opportunities

Even though there have been improvements in using EEG to decode silent speech and combining machine learning with brain-computer interfaces, there are still important issues that stop these technologies from being widely used, easily scaled up, and dependable. It's crucial to find and fix these problems to make BCIs useful in real-life situations, especially for people who can't speak.

1. **Limited Focus on Word-Type-Level Clustering**
   Most of the research on decoding silent speech so far has looked at tasks like recognizing individual sounds or classifying words as either one thing or another. These studies are useful, but they don't really show the full picture of how real language works. Fewer studies have looked into grouping speech based on types of words, like nouns, verbs, and other words that help build sentences. It's still unclear if these different types of words create different patterns in the brain when people imagine speaking.

2. **Data set Scarcity and Generalization Issues**
   A big problem in this area is the lack of large, open EEG data sets that have detailed language labels for silent speech. Most of the data sets that exist are limited in the number of words they cover, the variety of people they include, and the number of tests or trials they have. This lack of data makes it hard for machine learning models, especially deep learning models, to work well across different situations and not get stuck in patterns that only fit the data they were trained on.
   Also, EEG signals change a lot from one person to another and even within the same person over different sessions.
   This makes it hard to use models trained on one set of data on another person or another recording session. Even though some recent studies have looked into ways to adapt models to new data or transfer learning between different data sets, there is still no strong, reliable way to decode imagined speech without needing to train the model on each individual's data.

3. **Insufficient Use of Unsupervised and Self-Supervised Methods**
   Even though unsupervised learning is becoming more popular, its full power hasn't been used much yet. A lot of research still depends on supervised classification methods that need data with labels added by people. But labeling speech that's imagined is really hard and can be biased. There's a strong chance to use self-supervised learning and deep clustering methods more. These techniques can pick up useful patterns without needing exact labels, making them perfect for EEG work where there aren't many labeled examples.

4. **Limited Interpretability of Learned Representations**
   Another important issue is that many machine learning models, especially deep neural networks like auto-encoders and CNNs, are hard to understand. Even though these models work well, they often act like "black boxes," making it difficult to see how they process imagined speech. Creating models that can clearly show how hidden features relate to brain functions or language structures would improve both scientific knowledge and real-world use.

5. **Under-explored Evaluation Metrics for Clustering Quality**
   Clustering studies in EEG-based BCIs usually use simple measures like Silhouette Score or Purity. These measures don't fully show how meaningful the clusters are. There is a need for better ways to evaluate clusters that look at more than just how separate they are. They should also check how well the clusters match language, stay consistent across different sessions, and make sense from a cognitive point of view. Using semantic models or cognitive ontologies with clustering can make the evaluation more complete.

6. **Integration with Real-Time and Interactive Systems**
   Most studies are still limited to offline, proof-of-concept setups. Real-time use of silent speech BCIs is not often looked into, even though it's very important for helping people communicate. There's a lack of work on turning these systems into fast, interactive tools that can work well in messy, changing places like hospitals or homes. Looking into smaller models, learning that changes as needed, and getting feedback from users might help fill this gap.

Research Opportunities

To fix these issues, there are several good research areas to look into:

- Creating big, open datasets with EEG data for silent speech that include many different words, speakers, and languages.
- Looking at how EEG responses group together based on language features like meaning, sentence structure, and word types to better understand how the brain represents language.
- Using and improving deep learning methods that don’t need labels, like combining auto-encoders with clustering, contrastive learning, and graph-based methods that work with EEG data.
- Making models easier to understand by using explainable AI tools such as saliency maps, layer-wise relevance methods, or showing how hidden features relate to real data.
- Building real-time brain-computer interfaces that can adapt to each user’s brain signals with smart signal processing and personalized models.

3. METHODOLOGY

3.1. DATASET DESCRIPTION

The data used in this study came from a public EEG database that was created for experiments where people imagine speaking. The database has recordings from 20 healthy people, all between the ages of 20 and 35, who are native speakers and do not have any known brain disorders. The EEG signals were collected using a 64-channel cap that follows the international 10–20 system, and the machine recorded the signals at a rate of 1024 Hz. The participants were told to silently think about saying words from different types—like nouns, verbs, and function words—without moving their muscles or eyes much.

During the experiment, each word was shown on a screen, and then a signal told the person to start imagining the word.

Each time they did this, they had a break to rest and avoid getting tired. All the recordings were made in a room that was quiet and protected from outside electrical signals to make sure the brain

3.2. Preprocessing Techniques

To improve the quality of the signal and get ready for using the EEG data in feature extraction and modeling, several steps were taken during preprocessing:

- First, a band-pass filter was used to keep only the EEG frequencies between 1 and 40 Hz. This helped remove slow drifts and unwanted high-frequency noise.

- Next, Independent Component Analysis was applied to identify and remove parts of the signal that were caused by things like eye blinks, muscle movements, or electrical interference.

- Then, the data was split into short time segments. Each segment was taken around the moment when the person imagined a word, specifically from −0. 5 seconds to +2. 5 seconds after the cue. This helped focus on the brain activity related to silent speech.

- To make the data more consistent across different people and sessions, baseline correction was done, and the data was also normalized using z-scores.
- This helps in reducing differences between individuals and makes the data easier to work with.

3.3. Feature Extraction & Autoencoder Design

The preprocessed EEG data was turned into a set of features using a stacked auto-encoder. This auto-encoder did two things at once: it helped extract important features from the data and reduced the number of features to make it easier to work with. The structure of the autoencoder was as follows:

- Input layer: 64 EEG channels with 256 time points for each recording.
- Encoder: Three layers with fewer and fewer features (like 1024 → 512 → 128), and each layer used a ReLU function to process the data.
- Latent space: A middle layer with 32 features that captured the most important information from the data in a compressed form.
- Decoder: This part was like the encoder but in reverse, taking the compressed features and rebuilding the original input data.

The autoencoder was trained using a Mean Squared Error (MSE) loss function, and the Adam optimizer with a learning rate of 0. 001 was used to improve its performance. To stop training early and avoid overfitting, early stopping was applied along with dropout regularization, which randomly ignored some data points during training with a rate of 0. 3. The hidden codes created by the encoder were then taken and used as input for clustering.

3.4. Clustering Algorithms Used

To find natural groups in the EEG data based on different word types, three clustering methods were used:

- K-Means Clustering: This method was chosen because it's easy to use and works well for dividing data into k separate groups. It was run several times with k values from 2 to 10.
- DBSCAN: This technique helps find clusters of any shape by looking at how closely data points are packed together. It's good at handling noisy data and unusual patterns.
- Hierarchical Agglomerative Clustering: This method builds a tree-like structure of clusters, and the connections between them are shown in a dendrogram, which helps in understanding the grouping better.

All these clustering methods were applied to 32-dimensional features that came from an auto-encoder.

3.5. Evaluation Metrics

To check how good and clear the clusters are, both internal and external measures were used:

- Silhouette Score: This looks at how far apart the clusters are. A higher score, close to 1, means the clusters are well separated.

- Davies–Bouldin Index (DBI): This checks how tight the clusters are inside and how far apart they are from each other. A lower score means the clustering is better.

- Confusion Matrix: This was made by comparing the clustering results with the real word-type labels (these were not used during training). It helps see how well the clusters match the actual meaning groups.

4. RESULTS

4.1. Baseline Clustering on Raw EEG

The first step was to group the data right away from the cleaned EEG signals, without using any methods to simplify the data. The dataset had many channels with time-based data and was split into parts that matched the imagined words. Each part was then adjusted to match the same scale across different trials.

Using K-Means to group this raw data didn't work well.

The scores from the Silhouette test were all under 0. 25, which means the groups weren't clearly separated. DBSCAN also didn't find any clear clusters, instead marking a lot of the data as random noise because the data was too complex and hard to separate. Hierarchical clustering made tree-like charts with short branches, showing that the raw EEG data wasn't good at showing differences between groups. This shows that better ways of transforming the data are needed to find hidden patterns that aren't obvious in the original signals.

4.2. Post-Autoencoder Clustering Performance

After training the autoencoder on the processed EEG data, we got 32-dimensional representations that helped group the data better. These representations made it easier to find clear and separate groups.

- Using K-Means on these transformed data gave a Silhouette Score of 0.49 and a Davies–Bouldin Index of 0. 82, which shows the groups are more tightly packed and easier to tell apart.

- DBSCAN found natural groups that match different word types and had much fewer unusual points compared to when we didn't use the transformed data.

- Hierarchical clustering also worked better, creating clearer and more detailed tree-like structures, showing better organization of the groups.

These results show that the autoencoder successfully learned important features from EEG signals that help tell apart different patterns of silent speech.

4.3. PCA/t-SNE Visualizations

To better see how the data groups together and check if the hidden space is clearly separated, we used two methods to reduce the number of dimensions: Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE).

- The PCA results showed some clear separation between groups in the first two main components, which supported the clustering results after using the autoencoder.

- t-SNE created a 2D visual representation of the 32-dimensional hidden data points, showing clearly different groups that matched various word types like nouns, verbs, and function words. The separation in t-SNE was much clearer than in PCA, suggesting that the structure of the EEG data has a non-linear pattern.

These pictures showed that the autoencoder worked well in separating the brain activity patterns related to different types of words during imagined speech.

4.4. Word Type Separation in Clusters

To check how well the clusters matched different types of words, the real word labels (used just for checking accuracy) were compared with the clusters made by the algorithms. A confusion matrix was created for the results from K-Means and DBSCAN:

- Cluster A had mostly noun words, with more than 78% of its words being nouns.

- Cluster B mainly had function words, but it also had some verbs mixed in.

- Cluster C mostly had verb words (around 72% match), which suggests that the brain shows different activity patterns when imagining to say action-related words.

These results show that grouping words in the hidden feature space can find meaningful groups based on meaning, even without actually speaking.
This supports the idea that different types of words, because they have different meanings, grammar, and movement plans, create different patterns in brain activity when people imagine speaking.

5. DISCUSSION

5.1. Analysis of Cluster Behavior by Word Type

The results from grouping the data, which came from EEG signals that were compressed using an autoencoder, showed that the brain encodes different types of words in unique ways during imagined speech. The way the groups separated in PCA and t-SNE charts, along with better scores from the Silhouette Score and Davies–Bouldin Index, supports the idea that different kinds of words, like nouns, verbs, and function words, create different patterns of brain activity.

Cluster A mostly included nouns, which are linked to specific objects and images.

This might explain why the brain activity for nouns is stronger and more consistent. Cluster B had more function words, which are more about grammar and structure, not specific meanings. That may be why the brain activity for these words is less clear and spread out. Cluster C was mostly verbs, which are connected to actions and movement. Since thinking about speaking uses the same brain areas involved in planning movement, it makes sense that verbs, which usually describe actions, create specific patterns in the brain that relate both to movement and thinking. These patterns can be picked up by EEG signals.

These results show that the brain arranges inner speech not just based on sound patterns but also on meaning and grammar. This can be discovered using methods that learn without needing labels. The fact that grouping words into categories without any guidance supports the idea that we can understand mental speech in a clear and organized way.

5.2. Consistency Across Sessions/Subjects

Another important part of this study looked at how well the clustering results stayed the same across different sessions and people. It's a common problem in EEG systems that performance can vary between sessions because of things like where electrodes are placed, background noise, and how focused or relaxed a person is. Even with these issues, the method using an auto-encoder to extract features did pretty well in keeping results consistent.

- When they combined data from multiple sessions and did clustering, the purity of the clusters stayed pretty much the same, which means the hidden features learned still held onto important information about the task.

- Looking at results for each person showed that there were differences between people, especially in how they represented function words.
  This could be because people use different ways to think about language inside their heads.

- Even though there wasn't complete agreement between all subjects, the main patterns linked to nouns and verbs were similar across people. This supports the idea that these types of words create similar patterns in the brain, which may be universal.

Although the patterns didn't match exactly in every case, the findings suggest that there could be common brain signals for silent speech.
These signals might be more reliable if we use proper adjustments and train models across different people.

5.3. Limitations

Even though the results looked encouraging, there were some issues that made the study's findings less reliable and harder to apply more widely.

- First, there wasn't enough data. The number of people and tests was small, which makes it harder to draw solid conclusions from the clustering results. This also means that using more complex models like transformers might not work well, since they could start memorizing the data instead of learning from it.

- Second, the brain signals had a lot of background noise and unwanted signals. Even after using methods like ICA to clean things up, there were still some random disturbances from things like muscle movements or outside sounds. These leftover issues might have made it harder to see clear patterns in the data.

- Inter-subject variability: EEG signals vary a lot from person to person, and differences in how people think or imagine words add more noise. Even though some patterns were found, it's still hard to apply these results to others.

- Lack of real-time validation: This study was done after collecting data, not during real-time use. The models haven't been tested in actual brain-computer interface systems, which would need better timing, reliability, and ways to give users feedback.

These problems, while common in early studies, show where more work is needed for better results in future research.

5.4. Future Research Directions

Based on the ideas and limits found in this study, here are some ways to move forward with silent speech decoding using clustering and deep learning:

- Getting bigger and more varied data: Making the dataset include more people, different backgrounds, and a wider range of words can help the

model work better for different situations and make the results more reliable.
Using data that includes words from different languages or phrases might also help understand how the brain handles language better.

- Using transformer models: This study used autoencoders to find features, but using transformers and attention methods could help understand the timing and meaning of brain signals better.
  New tools like EEG-BERT show that these methods might be really useful for this kind of work.

- Cross-subject adaptation methods: Approaches like domain adaptation, transfer learning, and adversarial learning can help deal with differences between people by learning features that stay the same across individuals. This can make brain-computer interfaces (BCIs) more reliable when used in real life.

- Integration with real-time decoding systems: Future research should focus on linking these clustering methods with online BCI systems that offer immediate feedback. This would let the system be tested in real-world situations and allow it to learn and adjust based on how users interact with it.

- Exploration of additional modalities: Using EEG along with other methods like EMG (electromyography) or fNIRS (functional near-infrared spectroscopy) could boost accuracy by getting more types of neural signals.

6. CONCLUSION

This study looked at a new way to understand silent speech by using clustering methods on brain data. The main idea was that different kinds of words—like nouns, verbs, and function words—create unique patterns in the brain when someone imagines speaking. The research used a process that included cleaning the brain signals, finding important features with an autoencoder, and grouping similar patterns together. The goal was to see if these imagined words could be sorted into clear groups just by looking at how their brain activity looks.

The main goal of this study was to do two things: first, check if silent speech EEG data has enough useful information to group different types of words together; and second, see if using unsupervised learning, especially with deep neural networks, can help decode internal speech without needing labels. The study's results show that both goals were met. When clustering was done on the raw EEG data, the groups were not clear or distinct, which shows the challenges of working with high-dimensional and noisy data. But after using a stacked autoencoder to process the data, the clustering improved a lot. The new features found through this method allowed algorithms like K-Means, DBSCAN, and Hierarchical Clustering to find clear, separate groups that matched the actual word types well.

The analysis showed some important results:

- Using an autoencoder to reduce the number of features made the clustering more clear and easier to understand. Measures like the Silhouette Score (which reached up to 0. 49) and the Davies–Bouldin Index (which dropped to 0. 82) showed that the clusters were better separated in the reduced space compared to the original data.

- When looking at the data using PCA and t-SNE, it became clear that the brain activity patterns for imagined words don’t spread out randomly in the feature space.
- Instead, they group together based on meaning or grammar.

- Nouns and verbs made tighter groups, probably because they have stronger meanings and are connected to movement in the brain.
- Function words, which don’t have clear meanings or movement-related signals, spread out more in the data.

- Even though there were differences between people and across sessions, some brain patterns stayed consistent, especially for words with strong meanings or actions.

These results show that the way language is structured is also reflected in how the brain activates during imagined speech, even when no actual words are being spoken. This understanding is important for improving Brain-Computer Interfaces (BCIs) that help restore speech, especially for people who can't speak because of conditions like ALS, stroke, or surgery on the voice box.

Clustering plays a key role in BCIs that work with silent speech.

Most speech-related BCI systems use supervised learning, which needs a lot of labeled data for training. That process can take a lot of time and isn't always practical, especially in medical situations. This study shows that unsupervised learning can be a better option. Clustering helps group brain activity based on natural patterns without needing labels, making it a flexible and efficient way to build personalized BCI systems that can quickly adjust to new users or changes without needing a lot of setup.

This work brings several new ideas to the table:

- It goes beyond just recognizing individual sounds or simple yes/no answers to classify entire words, which is more useful for everyday communication.
- By combining autoencoders with clustering methods, it connects deep learning with the study of how the brain processes language, leading to better and easier-to-understand brain-computer interfaces.
- The research also introduces a new way to use the brain's hidden patterns to understand meaning, which could lead to more advanced assistive technologies that do more than just spell words or say yes or no.

In terms of impact, this method has the chance to change how we design tools that help people communicate. By connecting high-level language patterns with brain activity, future brain-computer interfaces (BCIs) could let users send full words or sentences just by thinking—bringing us closer to real conversation. These systems might make communication faster and more expressive, and they could also help reduce mental exhaustion by working with how the brain naturally handles language.

Looking ahead, this research opens up several exciting possibilities:

- First, gathering bigger and more varied EEG data that includes people who speak different languages, different mental states, and a wider range of words.
- Second, using transformer-based models to better understand the timing of EEG signals and make the system work well for more people.
- Third, applying techniques like cross-subject adaptation and transfer learning to deal with differences in how people's brains work.
- Finally, putting this clustering method into real-time BCI systems to test how quickly they respond, how well users adjust and how useful it is in real situations.

In conclusion, this study shows that there is a clear pattern in the brain activity recorded during silent speech, and that using clustering without labeled data along with deep learning can uncover these patterns. These results represent progress in creating more natural, smart, and useful brain-computer interfaces that help with communication. The research highlights how thinking can be used as a way to interact, allowing language to be understood not from speaking aloud, but directly from the mind.

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APPENDICES

1. AUTOENCODER MODEL SUMMARY

The autoencoder used in this study was designed to learn compressed representations of EEG signal epochs. The architecture is as follows:

- Input Layer: 64 EEG channels × 256 time points (flattened to 16,384 units)
- Dense Layer 1: 1024 units, ReLU activation
- Dense Layer 2: 512 units, ReLU activation
- Dense Layer 3: 128 units, ReLU activation
- Latent Layer (bottleneck): 32 units
- Dense Layer 4: 128 units, ReLU activation
- Dense Layer 5: 512 units, ReLU activation
- Dense Layer 6: 1024 units, ReLU activation
- Output Layer: 16,384 units (reshaped back to 64 × 256)

Training Configuration:

- Optimizer: Adam
- Learning Rate: 0.001
- Loss Function: Mean Squared Error (MSE)
- Regularization: Dropout (rate = 0.3)
- Early stopping based on validation loss.

2. RAW EEG SAMPLE

he following is a sample of raw EEG data recorded from two channels (F3 and C3):

Time (s)	Channel F3 (μV)	Channel C3 (μV)
0.00	2.14	1.85
0.01	2.17	1.80
0.02	2.10	1.82
0.03	2.08	1.78
0.04	2.12	1.79

3. CLUSTERING ALGORITHM CODE

```
# Required libraries
import numpy as np
from sklearn.cluster import KMeans, DBSCAN, AgglomerativeClustering
from sklearn.metrics import silhouette_score, davies_bouldin_score
import matplotlib.pyplot as plt
from scipy.cluster.hierarchy import linkage, dendrogram

# Assume latent_features is a 2D NumPy array from your autoencoder
# latent_features = encoder.predict(preprocessed_eeg_data)
# K-Means Clustering
kmeans = KMeans(n_clusters=3, random_state=42)
```

```
kmeans_labels = kmeans.fit_predict(latent_features)

# Evaluation
kmeans_silhouette = silhouette_score(latent_features, kmeans_labels)
kmeans_dbi = davies_bouldin_score(latent_features, kmeans_labels)

print("K-Means Clustering Results:")
print(f"Silhouette Score: {kmeans_silhouette:.3f}")
print(f"Davies–Bouldin Index: {kmeans_dbi:.3f}\n")

# DBSCAN Clustering
dbscan = DBSCAN(eps=0.5, min_samples=5)
dbscan_labels = dbscan.fit_predict(latent_features)

# Filter out noise for evaluation
mask = dbscan_labels != -1
if len(set(dbscan_labels[mask])) > 1:
   dbscan_silhouette = silhouette_score(latent_features[mask],
dbscan_labels[mask])
   dbscan_dbi = davies_bouldin_score(latent_features[mask],
dbscan_labels[mask])
   print("DBSCAN Clustering Results (excluding noise):")
   print(f"Silhouette Score: {dbscan_silhouette:.3f}")
   print(f"Davies–Bouldin Index: {dbscan_dbi:.3f}\n")
else:
   print("DBSCAN did not form valid clusters for scoring.\n")

# Hierarchical Clustering
hierarchical = AgglomerativeClustering(n_clusters=3, linkage='ward')
hier_labels = hierarchical.fit_predict(latent_features)

# Evaluation
hier_silhouette = silhouette_score(latent_features, hier_labels)
hier_dbi = davies_bouldin_score(latent_features, hier_labels)

print("Hierarchical Clustering Results:")
print(f"Silhouette Score: {hier_silhouette:.3f}")
print(f"Davies–Bouldin Index: {hier_dbi:.3f}")

# Dendrogram
linked = linkage(latent_features, method='ward')
plt.figure(figsize=(8, 5))
dendrogram(linked, truncate_mode='level', p=5)
plt.title("Hierarchical Clustering Dendrogram")
plt.xlabel("Sample Index")
plt.ylabel("Distance")
plt.tight_layout()
```

```
plt.show()
```

4. EVALUATION METRIC FORMULAE

1. Silhouette Score

The Silhouette Score shows how much an object fits in its own group compared to other groups. The score goes from -1, which means the grouping is not correct, to +1, which means the grouping is perfect.

$$S(i)=b(i)-a(i)/\max\{a(i),b(i)\}$$

Where:

a(i): The average distance between sample i and all other points in its own cluster.

b(i): The average distance between sample i and the closest points in other clusters.

The overall silhouette score is calculated by taking the average of all S(i) values for each sample.

2. Davies–Bouldin Index (DBI)

The DBI measures how good a clustering is by looking at how similar items are within the same group and how far apart different groups are. Smaller numbers mean the clustering is better.

$$DB = \frac{1}{n}\sum_{i=1}^{n} max_{j\neq i}\left(\frac{\sigma_i+\sigma_j}{d(c_i,c_j)}\right)$$

Where:

σ_i: The average distance between each point in cluster i and the center of that cluster

d(c_i, c_j): The distance between the centers of clusters i and j

n: The total number of clusters

3. Confusion Matrix

The confusion matrix compares the real labels, like known word types, with the labels that are predicted by the clustering process. This helps check how well the clustering matches the real data. It is usually organized as:

	Predicted Cluster A	Predicted Cluster B	Predicted Cluster C
True Noun	TP	FN	FN
True Verb	FP	TP	FN
True Function Word	FP	FN	TP

Where:

TP: True Positives (when the model correctly identifies something)

FP/FN: False Positives/Negatives (when the model incorrectly identifies or misses something)

Even though unsupervised methods don’t use labels during training, confusion matrices can still be used later to evaluate performance.
This is done by matching the predicted groups with the actual labels, using the highest overlap as a way to connect them.